

1. Find the cartesian equation of the tangent plane to the level surface

$$\mathcal{S} := \{(x, y, z) \in \mathbb{R}^3: x^2 + \sin(x + y) - 2z^2 = -1\}$$

at the point $(\pi, \frac{\pi}{2}, \frac{\pi}{\sqrt{2}})$.

2. Find **all** the critical points of the function $f(x, y) := \sin x + \sin y$ and determine their nature.
3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := -x^3 - x \cos(x + y) \quad \text{for every } (x, y) \in \mathbb{R}^2.$$

(a) Find **all** the critical points of the function f .

(b) Determine the nature of the critical points $(0, \frac{3\pi}{2})$, $(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} + \pi)$, $(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} + \pi)$.

4. Let $(a, b, c) \in \mathbb{R}^3$ be such that $a^2 + b^2 + c^2 \neq 1$. Find on the sphere $x^2 + y^2 + z^2 = 1$, the points which are the closest and the points which are the furthest to the point (a, b, c) .
Hint: It is important to get the right setting, i.e., finding the right function to be minimized and maximized subject to the sphere $x^2 + y^2 + z^2 = 1$.

5. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) := -\frac{1}{4}x^4 + x^2y^2 - 4x^2y \quad \forall (x, y) \in \mathbb{R}^2$.

(a) Find all the critical points of the function f and determine the nature of each point.

(b) Find the points of maximum and minimum of the function f subject to $4x^2 + y^2 - 4y = 0$.

6. Find the points of maximum and minimum of the function $f(x, y) = xye^{-xy}$ over the domain

$$D := \{(x, y) \in \mathbb{R}^2: 1 \leq x \leq 4, \quad 0 \leq xy \leq 1\}$$